CSCE 3701 · TEAM CLOSET · MILESTONE 2 FINAL

Test Plan and Report

Version 1.3 · revision 5 · application build v1.6

Team	Habiba · Salma · Bashar · Karim · Youssef
Date	10/29/2026
Supersedes	v1.2 (build v1.5). Every number below is from the final runs of build v1.6, stored in `evidence/v16/`. The v1.2 report is appended unchanged as history; its numbers are superseded.

R5.1 Results at a glance

599/599
tests passing · 0 failing

407 · 192
unit · API (real PostgreSQL 16)

97.53 %
line coverage · branches 83.64 % · functions 90.84 %

3/3
demo checks on fresh seeds

R5.2 How the tests were run (honest)

> **Environment limitation.** The npm registry was blocked in the environment used for this pass, so the real Jest, Supertest, Express, pg and Vite packages could **not** be installed. The suites ran under `node:test` with small compatibility shims for those packages, against a real PostgreSQL 16, and the web app was bundled with esbuild instead of Vite. `npm test` on a normal machine is the authoritative run. The GitHub Actions workflow is **configured, not executed** for this build.

R5.3 Per test file

File	Executed	Passed	Failed
`tests/unit/ask_closet_intents.test.js`	63	63	0
`tests/unit/assistant_weather.test.js`	34	34	0
`tests/unit/closetGaps.test.js`	7	7	0
`tests/unit/diversity.test.js`	16	16	0
`tests/unit/engine.test.js`	29	29	0
`tests/unit/final_flow.test.js`	8	8	0
`tests/unit/insights.test.js`	38	38	0
`tests/unit/intelligence.test.js`	23	23	0
`tests/unit/lifecycle.test.js`	30	30	0
`tests/unit/offline_parity.test.js`	11	11	0
`tests/unit/security.test.js`	18	18	0
`tests/unit/shopping_logic.test.js`	8	8	0
`tests/unit/statusMachine.test.js`	86	86	0
`tests/unit/validate.test.js`	36	36	0
`tests/api/acceptance.test.js`	15	15	0
`tests/api/auth.test.js`	18	18	0
`tests/api/availability.test.js`	10	10	0
`tests/api/final_flow_api.test.js`	5	5	0
`tests/api/intelligence_api.test.js`	21	21	0

File	Executed	Passed	Failed
`tests/api/lifecycle_api.test.js`	12	12	0
`tests/api/preview_features.test.js`	40	40	0
`tests/api/proactive_api.test.js`	22	22	0
`tests/api/profile_planner_privacy.test.js`	19	19	0
`tests/api/shopping_list_api.test.js`	12	12	0
`tests/api/wardrobe.test.js`	18	18	0

R5.4 Browser journeys (Playwright, Chromium)

Journey	Target	Result
demo_journey.py (presentation flow)	live app	58 steps, 0 browser errors
stateful_journey.py	live app	live app: 15 checkpoints, 0 browser errors
lifecycle_journey.py (care, lend, donate, pack, purchase, planner)	live app	live app: 12 checkpoints, 0 browser errors
stateful_journey.py	offline demo	offline demo: 15 checkpoints, 0 browser errors
lifecycle_journey.py	offline demo	offline demo: 12 checkpoints, 0 browser errors

lifecycle_journey.py was also run 5 times in a row after defect D-22 was fixed: 5/5 passed. Before the fix it failed 1 of about 8 runs.

R5.5 Demo recording

The 13-step walkthrough was re-recorded after the last correction on a fresh seed: **5:33 (332.7 s)**. Frames of the shopping, donation, Ask Closet and final Home steps were checked by eye; this review found D-19 and D-23 (both fixed, then re-recorded).

R5.6 New tests in v1.6

IDs	Technique	What they show
ST-01 – ST-06	State transition	all 81 ordered pairs of the 9 states: 21 allowed, 60 rejected with 409; loans / care jobs opened and closed with the status
LF-01 – LF-06	EP · decision table	lifecycle helpers; seeded closet; message wording (D-23)
LF-10 – LF-12	EP over prompts	22 v1.6 Ask Closet prompts route to the right function with the right arguments; donate only
LC-01 – LC-04	Scenario	exact Black blazer flow: Needs alteration → tailor → left out of looks → reminder → Mark collected
LC-10 – LC-12	Scenario · BVA	lend with dates, overdue, Planner conflict, return keeps history
LC-20 – LC-23, LC-30	Scenario	multi-day plan skips non-wearable states; purchase impact ignores Ready to donate; rescue with and without a formal piece
OF-10 – OF-11	Parity	care and lending on the offline demo

R5.7 Defects found in this pass

ID	Found by	Severity	Feature	Description	Fix	Status
D-18	Lifecycle E2E (lifecycle_journey.py, loan step) → LC-10	Medium	Borrow & lend, Care (v1.6)	The server's "today" was the UTC day while the seed and the browser used Cairo midnight. Near midnight a loan showed −1 days, the tailor "ready today" reminder did not fire and a laundry conflict was missed.	One day for the app: the UTC day (isoDay). The repository receives the service's today() for lent_on / returned_on / closed_on; the seed and the E2E journeys use UTC days too.	Fixed, re-tested
D-19	Demo video review (shopping step) → LC-22	Medium	Purchase simulation (FR-94)	"Check my closet" ignored the item name, so white loafers got "Skip" in the quick check while the simulation below said "Useful purchase", a visible contradiction. It also counted pieces waiting for donation as owned.	/shopping/check accepts `name` (≤ 40 chars) and compares only owned pieces (isOwned). The video shows Good buy for white loafers and Skip for the black fitted top.	Fixed, re-tested
D-20	LF-05 (unit)	Low	Still love it? (FR-81)	The forgotten-piece list offered the camel cardigan, which is waiting for a repair and cannot be worn.	Forgotten candidates and outfit suggestions use isWearable (Available only).	Fixed, re-tested
D-21	Stateful E2E (Ask "Is my blazer clean?")	Low	Ask Closet	With the new black blazer in the seed, "my blazer" became ambiguous and Ask Closet asked "Which blazer?". This was correct behaviour, but the journey and demo script assumed one blazer.	Journeys, demo check and video ask about the "navy blazer"; the ambiguity question is kept (correct behaviour).	Fixed (test data), re-tested
D-22	Lifecycle E2E, flaky 1 of ~8 runs → 5/5 after the fix	Medium	Donation from Home "Still love it?"	After Donate, the Home card re-fetched at once. The denim jacket was no longer a candidate, so the card, and the open dialog with "Mark as donated", could unmount before the user saw it (a race).	The Home and Analytics cards refresh only when the dialog closes.	Fixed, re-tested

ID	Found by	Severity	Feature	Description	Fix	Status
D-23	Demo video review (final Home frame) → LF-06, API-K03	Low	Availability conflict notices, Ask Closet	“Your black loafers is packed until 2026-11-02”: wrong verb for plural garments and a raw ISO date in user-facing text. API-K03 had asserted the wrong wording.	isAre() and onDay() in shared/lifecycle: “Your black loafers are packed until Mon 2 Nov.” Also used for loan, tailor and borrow messages. API-K03 corrected.	Fixed, re-tested
D-24	LF-10 (unit, prompt classes)	Low	Ask Closet care questions	“Anything that needs an alteration?” was not understood (the pattern did not allow “an”).	The care pattern accepts “a / an”.	Fixed, re-tested
D-25	Lifecycle E2E (screenshot review)	Low	Status badges	Statuses with spaces (“Needs repair”, “At tailor”) produced two CSS classes, so the badge lost its colour.	statusClass() turns spaces into dashes.	Fixed, re-tested

R5.8 Not verified

- The real npm packages, Vite build and GitHub Actions run (see R5.2).
- The interactive PowerPoint deck’s playback.
- Deployment: Render blueprint configured; the local demo is the verified presentation path.
- Jira: proposed stories US-112 – US-117 are not in Jira; no screenshots were produced.

> The superseded v1.2 report (build v1.5) follows as history.

CSCE 3701 SOFTWARE ENGINEERING · THE AMERICAN UNIVERSITY IN CAIRO

Test Plan and Test Report for Closet

Version 1.2 · Milestone 2 final · application build v1.5

Prepared by	Habiba Khashaba, Salma, Bashar, Karim and Youssef (Team Closet)
Instructor	Dr. Sherif Aly
Date	10/28/2026
Supersedes	Closet_M2_Test_Plan_and_Report.pdf (v1.1, build v1.4). Every number below is from the final run of build v1.5 and is reproduced in `05_Testing/evidence/`.

1. Summary

497/497
automated tests passing · 96 suites · 0 failing

317 + 180
unit tests + API tests (API tests on real PostgreSQL 16)

97.24%
line coverage · branches 83.42% · functions 91.09%

58 + 15 + 15
browser steps: demo journey · stateful (live) · stateful (offline demo)

How the final run was executed — read this first. The run was made in the team's build environment, where the npm registry was not reachable. The real server code, the real tests and a real PostgreSQL 16 database were used; nine npm packages were replaced by small *compatibility shims* (express, pg, bcryptjs, jsonwebtoken, helmet, cors, express-rate-limit, multer, dotenv; supertest for the API tests) and Jest by a thin adapter over Node's built-in test runner. The shims never ship. **Pending:** the authoritative `npm test` and `npm run test:coverage` with the real packages on a team machine, and the first execution of the GitHub Actions workflow (configured, **not yet executed**). Results of that run replace section 4 if they differ.

Exit criterion (section 2.6)	Result
All Sprint 1 acceptance tests AT-01 – AT-15 pass	met 15/15
No open High or Medium defect	met 17 found, 17 fixed and re-tested
Line coverage ≥ 80 % on server modules	met 97.24 % (V8, files loaded by the tests)
Every new requirement FR-81 – FR-101 has at least one test	met see the traceability matrix v1.5
Demo journey and stateful journeys pass with 0 browser errors	met live and offline
Official npm run + CI execution	pending configured, not executed

2. Test plan

2.1 Scope

In scope: Sprint 1 stories (accounts, profile, add / browse / edit pieces, status and laundry, planner, gaps, delete account), the Sprint 2/3 previews built early (recommendations, change one piece, try-on preview, Mix & Match, Ask Closet, shopping list, learning) and the v1.5 proactive features FR-81 – FR-101. Out of scope: load testing, real third-party weather/LLM availability (adapters are tested with fakes and the estimate mode), mobile browsers other than Chromium, and the Render deployment (blueprint configured; the local demo is the verified presentation path).

2.2 Levels and tools

Level	What	Tool	Where
Unit	pure rules: status machine, validation, engine, diversity, insights, parser, policies	Jest API (node:test adapter in the final run)	server/tests/unit
API / integration	routes through the full pipeline on a real database, including a second user for ownership checks	supertest + PostgreSQL 16	server/tests/api
Acceptance	AT-01 – AT-15 (Sprint 1 Definition of Done) and the API-PR proactive scenarios	same	tests/api/acceptance, proactive_api
End-to-end (browser)	demo journey (58 steps) and stateful journey (15 checkpoints)	Playwright (Python), Chromium	e2e/
Offline demo parity	the offline file runs the same services and rules	OF-01 – OF-09 + stateful journey on the file	tests/unit/offline_parity, e2e
Security	headers, CORS, auth, lockout, tokens, ownership (404), validation, photo privacy	unit + API	security.test, auth.test, profile_planner_privacy
Regression	one or more tests per defect D-01 – D-17; whole suite on every change	same	section 6

2.3 Test design techniques (black-box)

Technique	Applied to	Example test IDs
Equivalence partitioning (EP)	email, piece details, profile, events, Ask phrasings (56 questions in 12 intent classes)	UT-V01 · UT-V02 · UT-V03 · UT-V04 · UT-V05 · AI-43 …
Boundary value analysis (BVA)	password 7/8, lockout 4/5, photo 10 MB / +1 byte, weather 18/19 and 25/26 °C, recycle N days, gaps 1/2, unworn 89/90 days, monitor 28/30 °C	API-A08 · API-W07 · API-W08 · RE-05 · RE-06 · RE-07 · RE-08 · RE-22 · UT-G03 · UT-G04 · IN-01 · API-PR31
State transition (ST)	item status machine (28 tests: every valid and invalid transition), rehoming intention, learning card closed → flipped → confirmed → undone	Item status state machine suite · API-S01 – API-S06 · API-PR02 · API-PR03

Technique	Applied to	Example test IDs
Decision table	deletion reason → effect (replacement / learning / none) incl. Donated, Given away, Sold	Deletion reasons suite · API-PR02 · API-PR92
Scenario / use case	core flow, donate flow, monitor → adjust, trip capsule, Ask follow-ups	AT-01 – AT-15 · API-PR90 – API-PR93 · stateful journey
Error guessing	questions that look like commands (D-09), double save (D-10), duplicate notices (D-17)	AI-43 · API-PR61 · API-PR90

2.4 Environment

Node 22, PostgreSQL 16 (database `closet_test`; tables truncated between tests by `tests/helpers.js`), TZ Africa/Cairo, WEATHER_MODE=estimate (deterministic forecasts), seed from `server/src/db/seedData.js`. Browser tests: Chromium 1440 × 900, the app on `localhost:4000` after `npm run seed`, and the offline file `Closet_Demo.html`.

2.5 Entry criteria

- Migrations 001 – 006 apply on an empty database.
- Seed completes; `npm run demo:check` passes.
- Story has acceptance criteria in Jira.

2.6 Exit criteria

As in the table in section 1. A failing test blocks the merge (CI job `test`, configured).

2.7 Risks

Risk	Mitigation
Shimmed packages behave differently from the real ones	Shims implement only the used API; official npm run pending (section 1)
Weather provider unavailable on stage	estimate mode + offline demo
Random ids change the demo order	deterministic tie-breaks (D-07), demo-check over 3 fresh seeds
Thresholds (90 days, 5 °C) are guesses	BVA tests pin the current rule; values are constants to tune

3. Black-box test cases (selected, with inputs and expected results)

ID	Technique	Input	Expected	Result
API-A04	BVA	password of 7 characters	400 with field details	Pass
API-A08 / A09	BVA	4 then 5 failed logins	4th still allowed; 5th locks, 423	Pass
API-W07 / W08	BVA	photo 10 MB / 10 MB + 1 byte	accepted / 413	Pass
RE-05 – RE-08	BVA	25, 26, 18, 19 °C	warm allowed / none / layer required / not required	Pass
IN-01	BVA	piece unworn 90 / 89 days	candidate / not a candidate	Pass
API-PR31	BVA	planned at 24 °C, forecast 30 °C / 28 °C	weather alert on the blazer / no alert	Pass
API-S02	ST	Laundry → Borrowed	409, status unchanged	Pass
AI-43	EP	“Is my blazer clean?”, “Are my loafers available?”	read-only availability, state unchanged	Pass
API-PR61	Error guessing	Save then Favourite the same look	one saved record, favourite = true	Pass
API-PR02	Decision table	Donate → Mark as donated	piece gone, history kept, no Shopping Bag item	Pass
API-PR03	Security / EP	rehome another user’s piece; method “burn”	404; 400	Pass
DV (diversity suite)	BVA on overlap	seeded closet, 3 occasions	any two looks share ≤ 1 major garment	Pass

4. Results by test file (final run)

File	Tests	Passed	Failed
`tests/unit/ask_closet_intents.test.js`	63	63	0
`tests/unit/assistant_weather.test.js`	34	34	0
`tests/unit/closetGaps.test.js`	7	7	0
`tests/unit/diversity.test.js`	16	16	0
`tests/unit/engine.test.js`	29	29	0
`tests/unit/final_flow.test.js`	8	8	0
`tests/unit/insights.test.js`	38	38	0
`tests/unit/intelligence.test.js`	23	23	0
`tests/unit/offline_parity.test.js`	9	9	0
`tests/unit/security.test.js`	18	18	0
`tests/unit/shopping_logic.test.js`	8	8	0
`tests/unit/statusMachine.test.js`	28	28	0

File	Tests	Passed	Failed
`tests/unit/validate.test.js`	36	36	0
`tests/api/acceptance.test.js`	15	15	0
`tests/api/auth.test.js`	18	18	0
`tests/api/availability.test.js`	10	10	0
`tests/api/final_flow_api.test.js`	5	5	0
`tests/api/intelligence_api.test.js`	21	21	0
`tests/api/preview_features.test.js`	40	40	0
`tests/api/proactive_api.test.js`	22	22	0
`tests/api/profile_planner_privacy.test.js`	19	19	0
`tests/api/shopping_list_api.test.js`	12	12	0
`tests/api/wardrobe.test.js`	18	18	0
Total	**497**	**497**	**0**

Run summary from the log: tests 497, suites 96, pass 497, fail 0, duration 20.9 s. Log: `evidence/test_run_v15.txt`.

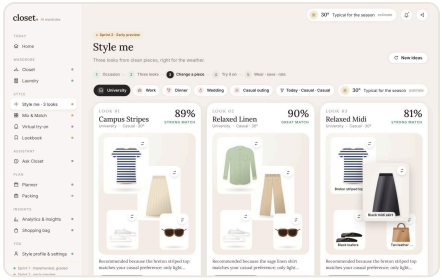
5. End-to-end and offline demo

Demo journey (`e2e/demo_journey.py`): 58 steps, 0 browser errors. **Stateful journey** (`e2e/stateful_journey.py`) — after every change it navigates away and back (and sometimes reloads) and checks the state through both the UI and the API:

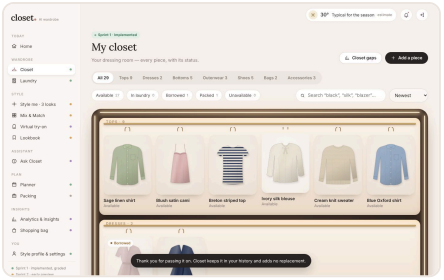
- Live app: live app: 15 checkpoints, 0 browser errors
- Offline demo file: offline demo: 15 checkpoints, 0 browser errors

#	Checkpoint (same assertions live and offline)
s01	Sign in; Home shows ≤ 2 proactive notices and “Still love it?” (denim jacket, 104 days)
s02	Navy blazer → In laundry; Laundry basket, Closet card and Home count follow; the blazer is not recommended
s03	Wash done → Available in the Closet and the API; basket empty
s04	Three looks; any two share ≤ 1 major garment (measured from the API)
s05	Favourite + Save → one record; still in Favourites after a reload; unfavourite keeps it in All looks
s06	Still love it? → Style it: the denim jacket is in all three looks
s07	Keep it (“Still love it”) → gone from Analytics and from the API candidates
s08	Donate → the piece is marked Ready to donate
s09	Mark as donated → gone from the closet, history kept, nothing added to the Shopping Bag
s10	Ask Closet: 12 questions → 12 different answers; blazer answer follows its status; “Add this” reaches the Shopping Bag; out-of-scope reply
s11	Mix & Match: red cropped tee lowers the score, Suggest a fix raises it (live run: 98 → 82 → 98)
s12	Remove the silk scarf with reason “Torn / damaged” → Replacement in the Shopping Bag
s13	Closet has learned: closed card → flip → Confirm changes the profile → Undo restores it
s14	Planner: planned at 24 °C, now 30 °C → Adjust outfit changes exactly one piece, the warm one
s15	Packing: carry-on capsule within 8 clothing / 2 shoes; back to Home

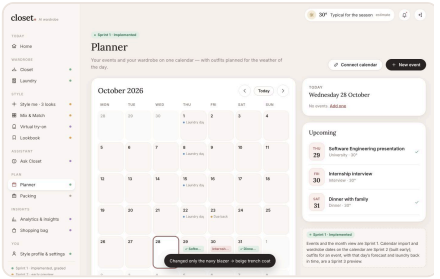
Screenshots of every checkpoint: `evidence/e2e-stateful-live/`, `evidence/e2e-stateful-offline/`. Demo preflight: `evidence/demo_check.txt` (deterministic over 3 fresh seeds).



s04_three_looks



s09_donated



s14_adjusted

6. Defect log and regression tests

ID	Found by	Severity	Feature	Description	Fix	Status
D-01	UT-V10 (unit, EP)	Low	US-01 register/login	An email typed with spaces around it (" Salma@Example.com ") was rejected as invalid instead of being trimmed.	Validator now trims before checking the format (validate.js).	Fixed, re-tested
D-02	API-W07 (API, BVA)	Medium	US-08 add a piece by photo	A photo of exactly 10 MB was refused with 413, although the acceptance criterion allows "≤ 10 MB". The upload library treats its limit as exclusive.	Upload limit set to 10 MB + 1 byte; API-W07 (10 MB) passes and API-W08 (10 MB + 1 byte) is still refused.	Fixed, re-tested
D-03	Integration test (browser journey, step "Add a piece")	Medium	US-08 add a piece by photo	The photo preview before saving was blank: the Content Security Policy from the security headers blocked blob: images. Found by running the full journey in Chromium.	CSP img-src now allows blob:. The Google sign-in script and popup are allowed explicitly, nothing else.	Fixed, re-tested
D-04	Integration test (repeated browser journeys)	High	All stories (availability of the API)	The general rate limit (300 requests / 15 min) also counted photo downloads. Browsing a 16-piece closet a few times used it up and every request returned 429 "Too many requests", which would have stopped the live demo.	Photo URLs are excluded from the general limit (they are already protected by signed URLs) and the limit is 1,000 / 15 min. Sign-in keeps its strict limit of 30 / 15 min.	Fixed, re-tested
D-05	API-A06 (API, preview suite)	Medium	Ask Closet “change the shoes” (Sprint 3 preview)	In an outfit with a dress there is no bottom, so “the shoes” were the 2nd piece, but the agent changed the 3rd piece by position — the bag. Found when a ranking change made a dress outfit come first.	The slot is now found by category (top or dress / bottom or layer / shoes / bag or accessory); if the outfit has none, the reply says so.	Fixed, re-tested

ID	Found by	Severity	Feature	Description	Fix	Status
D-06	Demo rehearsal (API call, final pass) → regression tests RE-31, API-FP01	High	US-59 Style this piece (Sprint 3 preview)	“Style this piece” on a forgotten Camel cardigan at 30 °C returned three looks **without the cardigan**: the weather filter removed the warm piece before the fixed-piece step, so the promise “every look contains it” broke.	The chosen piece is put back into the pool when it is Available (weather and recycle time do not apply to it); a fixed outerwear piece can now complete a top + bottom look; a piece that is not Available returns no looks and a plain note.	Fixed, re-tested
D-07	Demo rehearsal (3 runs from a fresh seed, demo:check) → regression test RE-34	Medium	US-58 Replace one piece (Sprint 3 preview)	When two alternatives in the replace tray had the same score (Sage linen shirt 99 %, Blue Oxford shirt 99 %), their order came from the random database ids, so every reseed showed a different order and the rehearsed demo could change on stage.	Ties are broken by piece name, then id (looks, tray, most-worn list). Three fresh-seed runs now give identical looks, tray, forgotten-piece looks, Mix & Match scores and Ask replies.	Fixed, re-tested
D-08	Code review (final pass, honesty audit)	Medium	US-20 save a look	After saving a look, the toast offered an "Undo" button whose action did nothing (`run: () ⇒ {}`), so the user was promised an undo that did not exist.	The fake Undo was removed; saving is now idempotent (D-10) so a second tap cannot create a duplicate.	Fixed, re-tested
D-09	Ask Closet intent matrix AI-40 (unit, EP) — reproduced on the v1.4 baseline	High	US-41 Ask Closet	"Is my blazer clean?" was classified as set_status and **changed** the blazer's status. A question changed state.	Questions are routed to a read-only availability intent; only imperative phrases ("put … in the wash") change state. AI-40 – AI-47 cover 56 phrasings.	Fixed, re-tested
D-10	Stateful E2E journey (Lookbook) → API-PR61	Medium	US-20 save / favourite	Save then Favourite (or Favourite then Save) created two saved records of the same look.	One saved look per set of pieces: POST /outfits returns the existing record (200) and merges the favourite flag.	Fixed, re-tested
D-11	Diversity tests DV-01 – DV-07 (unit, BVA on overlap)	High	US-19 three looks	The three looks often shared 2–3 major garments (same top and bottom, different shoes) — they were not three styling directions.	Diversity-aware selection (pickDiverse): any two looks share at most one major garment when the closet allows; relaxes step by step only when it cannot.	Fixed, re-tested

ID	Found by	Severity	Feature	Description	Fix	Status
D-12	Code review → API-PR (vocabulary)	Medium	US-30 change status	The web app had its own copy of the allowed status transitions, which had drifted from the server rules.	/vocabulary returns STATUS_TRANSITIONS; the Piece screen reads them from there (one source of truth).	Fixed, re-tested
D-13	Honesty audit of the offline demo → OF-01 – OF-09	High	Demo (all stories)	The offline demo replayed recorded responses: a change (e.g. laundry) did not propagate to other screens.	The offline demo now runs the real services on in-memory repositories (same seed, same rules); state persists for the browser session; Reset restores the seed.	Fixed, re-tested
D-14	Browser journey (demo_journey.py, Planner step)	Medium	Planned-outfit monitor	After "Adjust outfit" fixed the planned look, the weather alert stayed on the Planner until a reload.	The Planner reloads the monitor after a fix.	Fixed, re-tested
D-15	API-PR (purchase impact) — BVA on owned pieces	Medium	US-88 purchase check	Purchase analysis only counted Available pieces, so a closet with items in the laundry looked emptier and "new combinations" were overstated.	Every owned piece (not being rehomed) counts; only distinct major-garment sets are counted.	Fixed, re-tested
D-16	Ask Closet intent matrix (unit)	Low	US-41 Ask Closet	"Who has my rose wrap dress?" did not find the piece because "rose" was not a known colour word.	Colour vocabulary extended (rose, blush, ivory, sage) and piece names are matched word by word.	Fixed, re-tested
D-17	API-PR (notifications)	Low	Notification Center	An overdue borrowed piece produced two notices (due reminder + overdue borrow).	Due reminders are de-duplicated against overdue borrow notices; every notice has a stable key.	Fixed, re-tested

7. Security testing

Concern	Tests	Result
Security headers, no server banner, CORS allow-list, safe errors	API-X01 · API-X02 · API-X03 · API-X04 · API-X05	Pass
Passwords hashed (bcrypt), length rules, lockout after 5 failures	API-A02 · API-A04 · API-A08 – A10 · Passwords (bcrypt) suite	Pass
Tokens: missing, tampered, wrong secret, expired, after logout, deleted user	API-A12 – API-A15 · Login tokens (JWT) suite	Pass
Ownership: another user's pieces, photos, plans, trips, new v1.5 routes → 404	Ownership and photo privacy suite · API-L04 · API-PR01 · API-PR03	Pass
Validation on new fields (price, rehome method, days ranges)	API-PR03 · API-PR50	Pass
Admin sees no wardrobe data	Admin suites (US-80, US-81)	Pass
Offline demo requires its local session	OF-02	Pass

8. Coverage

V8 coverage of the server files loaded by the tests: lines 97.24 %, branches 83.42 %, functions 91.09 %. Files never loaded by a test do not appear in this report — the client (React) is covered by the browser journeys, not by line coverage. Per-file table: Appendix B.

9. Limitations (stated plainly)

- Final run used compatibility shims for npm packages (section 1); the official `npm test` on a team machine is pending.
- CI workflow (unit, API, E2E, offline demo jobs) is configured but has not been executed.
- Client production build with Vite not yet run (the offline demo was bundled with esbuild).
- No usability study with real users yet; no load tests; Chromium only.
- Try-On is a simulated overlay preview, tested as such (placement tests), not image generation.
- Render deployment: blueprint configured; the local demo is the verified presentation path.

Appendix A · All 497 automated tests of the final run

Sprint 1 acceptance: "A working closet" (15)

Test	Result
AT-01 US-01 Register and log in — password ≥ 8 chars; wrong password shows an error; no other user's data is returned	Pass
AT-02 US-02 Style quiz — ≤ 4 steps, skippable; answers saved	Pass
AT-03 US-03 Dress code and modest preference — one of the 5 levels; modest stored	Pass
AT-04 US-04 Set my city — saved to the profile (forecast shown from Sprint 2)	Pass
AT-05 US-08 Add a piece by photo — JPG/PNG/HEIC ≤ 10 MB stored and shown as the piece's image	Pass
AT-06 US-09 Enter details — values from 3.1; category, colour and ≥ 1 dress-code level required	Pass
AT-07 US-11 Review before saving — the piece exists only after Save	Pass
AT-08 US-12 Browse and filter — filters combine; counts update	Pass
AT-09 US-71 Edit a piece later — changes appear in filters	Pass
AT-10 US-56 Status per piece — exactly one status, shown as a badge	Pass
AT-11 US-15 Laundry basket — one tap each way; basket lists all Laundry pieces	Pass
AT-12 US-16 Wash done — all Laundry pieces become Available	Pass
AT-13 US-46 Add events by hand — title, date, time and one occasion; month view	Pass
AT-14 US-64 Closet gaps — a gap is ≤ 1 piece at a level the user dresses for	Pass
AT-15 US-06 Delete my account — double confirmation; all data and photos deleted	Pass

Registration (5)

Test	Result
API-A01 valid email and password → 201, token, profile created	Pass
API-A02 password stored as bcrypt hash, never in plain text	Pass
API-A03 duplicate email (any letter case) → 409	Pass
API-A04 7-character password → 400 with field details	Pass
API-A05 missing fields → 400 listing both	Pass

Login (6)

Test	Result
API-A06 correct credentials → 200 and a working token	Pass
API-A07 wrong password and unknown email give the same 401 message	Pass
API-A08 4 failures still allow login (boundary below lockout)	Pass
API-A09 5th failure locks the account; even the right password gets 423	Pass
API-A10 after the 15-minute lock expires, the right password works again	Pass

Test	Result
API-A11 suspended account cannot sign in (403)	Pass

Sessions and protected routes (4)

Test	Result
API-A12 no token → 401 on every protected route	Pass
API-A13 tampered, wrong-secret and expired tokens → 401	Pass
API-A14 logout ends the session: the old token stops working	Pass
API-A15 a token for a deleted user is refused	Pass

Google sign-in (US-01 "or Google") (3)

Test	Result
API-A16 first Google sign-in creates the account and profile	Pass
API-A17 Google sign-in links to an existing email account	Pass
API-A18 invalid Google credential → 401	Pass

Status per piece (US-56) (6)

Test	Result
API-S01 valid path Available → Borrowed → Laundry → Available	Pass
API-S02 invalid transition Laundry → Borrowed → 409 and status unchanged	Pass
API-S03 invalid transition Unavailable → Packed → 409	Pass
API-S04 setting the same status is a no-op (exactly one status, no duplicate history)	Pass
API-S05 unknown status value → 400	Pass
API-S06 the status badge value comes back with the piece	Pass

Laundry basket and Wash done (US-15, US-16) (4)

Test	Result
API-L01 one tap in, basket lists it; one tap out	Pass
API-L02 Wash done returns every Laundry piece to Available, and only those	Pass
API-L03 Wash done with an empty basket returns 0	Pass
API-L04 Wash done never touches another user's laundry	Pass

Three looks, change one piece (US-19, US-24) (2)

Test	Result
API-CF03 the three looks each have their own editorial name	Pass
API-CF04 the replace tray: up to 3 Available alternatives, each with its re-scored look and only that slot changed	Pass

Style a forgotten piece (US-54, US-59) (2)

Test	Result
API-FP01 a piece not worn for 50 days is "forgotten"; Style this piece gives 3 looks that all contain it — even a warm layer at 30 °C	Pass
API-FP02 a forgotten piece that is in the laundry cannot be styled: no looks, a plain note	Pass

Mix & Match and Suggest a fix (US-42, US-43, US-66) (1)

Test	Result
API-MM01 a strong look scores high; a weak piece drops the score; the best single swap (Suggest a fix) raises it again	Pass

Explainable recommendations (US-57) (2)

Test	Result
API-EX01 each recommended outfit explains why, using the profile and the weather	Pass
API-EX02 rejections feed the explanation ("Your feedback")	Pass

Wardrobe Health (US-89) (7)

Test	Result
API-H01 score with parts, penalties and actions; parts add up to the score	Pass
API-H02 a new user with no pieces gets an empty score, not an error	Pass
WH-01 empty wardrobe: score 0, flagged empty, one action	Pass
WH-02 perfect closet reaches exactly 100 (every part at its maximum, no penalties)	Pass
WH-03 the score is the sum of the parts minus the penalties (formula is transparent)	Pass
WH-04 penalties are capped at −10 each and the score never goes below 0	Pass
WH-05 potential duplicates: same category and colour with a shared style	Pass

Closet has learned (US-40) (7)

Test	Result
API-LR01 confirm a pattern → profile changes; undo → restored exactly	Pass
API-LR02 "Not accurate" dismisses without changing the profile; unknown pattern → 404	Pass
LR-01 two "Too formal" rejections → lower usual dress code, with evidence and confidence	Pass
LR-02 one rejection is not enough evidence	Pass
LR-03 favourite looks reveal a style the profile lacks	Pass
LR-04 neutral wear history adds the most-worn missing neutral	Pass
LR-05 rarely worn colour; previousOf keeps the exact values for Undo	Pass

Conflict detection (US-79) (3)

Test	Result
API-K01 a planned piece moved to Laundry (not back in time) creates a conflict with a replacement	Pass

Test	Result
API-K02 fix swaps only that piece; a piece back before the event is not a conflict	Pass
API-K03 a Packed planned piece is a conflict, and it shows in the Notification Center	Pass

Conflict detection: double booking (US-79) (1)

Test	Result
API-K04 the same piece planned for two events < 4 h apart on one day is flagged	Pass

Outfit history (US-91) (1)

Test	Result
API-OH01 wear with occasion and weather; filter by occasion and rating	Pass

Packing mode (US-68) (3)

Test	Result
API-PK01 plan proposes outfits per day, counts, and never a Laundry piece	Pass
API-PK02 save → mark as packed: pieces become Packed with a return date and leave recommendations	Pass
API-PK03 validation: end before start, too long, unknown trip, another user's trip	Pass

Notification Center (US-92) and activity log (US-93) (3)

Test	Result
API-N01 event tomorrow without an outfit → a notice; mark read	Pass
API-N02 Wash done creates a notice and activity entries	Pass
API-N03 the activity log records sign-in, profile update and a deleted piece by name, only for its owner	Pass

Admin system health (US-81) and templates (US-90) (2)

Test	Result
API-AD05 admin sees service health and config; a normal user gets 403	Pass
API-OT01 templates are part of the shared vocabulary	Pass

Duplicate purchase warning (US-65) (4)

Test	Result
API-DU01 two similar white tops → "You already own 2 similar white tops" with match details	Pass
API-DU02 a dissimilar piece does not trigger the warning	Pass
DU-01 a near-identical piece scores ≥ 70 and lists what matched	Pass
DU-02 a different category never counts; a different colour falls below the threshold	Pass

Recommendations (Sprint 2 early start) (6)

Test	Result
API-R01 three explained outfits with title, compatibility and breakdown	Pass

Test	Result
API-R02 occasion is required	Pass
API-R03 a bag counts as the fourth piece	Pass
API-R04 change one piece keeps the others	Pass
API-R05 Mix & Match check: graded breakdown and an explanation	Pass
API-R06 another user's piece cannot be styled (404)	Pass

Version 1.2 customization (review comments) (6)

Test	Result
API-C01 settings: laundry basket time, recycle time, laundry day, name	Pass
API-C02 laundry basket time: a piece comes back on its own after N days	Pass
API-C03 auto-return off: the piece stays in the basket	Pass
API-C04 recycle time: clothes worn yesterday wait; shoes may repeat	Pass
API-C05 borrowed with a return date and a note	Pass
API-C06 planning ahead: a piece in the laundry counts if it is back by the date	Pass

Delete with a reason (US-13) (3)

Test	Result
API-DL01 reason is required and must be one of the list	Pass
API-DL02 the piece and its photo are gone, the reason is kept	Pass
API-DL03 another user cannot delete my piece	Pass

Outfits, wear history and insights (4)

Test	Result
API-O01 save, favourite, list and delete	Pass
API-O02 wear → counts, last worn, analytics; unavailable piece refused	Pass
API-O03 reject with a reason; hide a piece for 7 days	Pass
API-O04 forgotten pieces: worn before, not for 40+ days	Pass

Ask Closet (Sprint 3 early start) (10)

Test	Result
API-AC01 "What should I wear to university tomorrow?" → outfits	Pass
API-AC02 "Do I own something similar to a black blazer?" → my blazer	Pass
API-AC03 "I want to buy black trousers. Do I need them?" → Skip when owned twice	Pass
API-AC04 put in the laundry, then undo	Pass
API-AC06 change one piece of the last outfit, keeping the rest	Pass
API-AC07 wash done, weather, help	Pass

Test	Result
API-AC08 hide a piece and undo; ambiguous piece asks which one	Pass
API-AC09 plan my outfit for the next event	Pass
API-AC10 bad context and bad undo id are refused	Pass
API-AC05 “What is missing from my wardrobe?” → shopping suggestions	Pass

Planner: calendar, reminders, import (3)

Test	Result
API-PL01 calendar markers: laundry day and a borrowed piece due back	Pass
API-PL02 reminders: plan an outfit, a planned piece in the laundry, a borrowed piece due	Pass
API-PL03 .ics import: future events only, occasion guessed, no duplicates	Pass

Smart Shopping (Sprint 3 early start) (4)

Test	Result
API-SH01 suggestions name a reason and what they pair with	Pass
API-SH02 a torn piece goes onto the shopping list as a replacement (not a duplicate suggestion)	Pass
API-SH03 dismiss hides a suggestion; bad key 404	Pass
API-SH04 “Do I need it?”: good buy vs. skip	Pass

Admin (Sprint 3 early start, US-80) (4)

Test	Result
API-AD01 a normal user gets 403	Pass
API-AD02 admin sees accounts with counts only, never pieces	Pass
API-AD03 suspend ends the user’s session at once; restore lets them sign in again	Pass
API-AD04 an admin cannot suspend themselves	Pass

Long-unworn pieces → style, keep or donate (user decides) (3)

Test	Result
API-PR01 the 104-day denim jacket is a candidate; Keep it snoozes it; another user cannot keep my piece	Pass
API-PR02 Donate: Ready to donate → left out of outfits → Mark as donated → gone, history kept, NO replacement	Pass
API-PR03 Cancel brings the piece back to normal; invalid method → 400; someone else’s piece → 404	Pass

Seasonal rotation (1)

Test	Result
API-PR10 at 30 °C the winter knit is suggested; Pack for season → Packed (seasonal storage); Bring back → Available	Pass

Laundry intelligence (6)

Test	Result
API-PR20 the two most-worn work tops in the laundry before a Work event → warning; Wash done clears it	Pass
API-PR21 a 3-day plan never repeats a top, bottom or dress and skips laundry pieces	Pass
IN-20 high-use Work tops in the laundry before a Work event → one warning	Pass
IN-21 irrelevant laundry (a casual piece, or no event) does not warn	Pass
IN-22 Wash done removes the warning	Pass
IN-23 a multi-day plan never reuses a top, bottom or dress before a wash; allowRepeats respects the user	Pass

Planned-outfit monitor: weather + availability, one system (3)

Test	Result
API-PR30 a planned piece goes to the laundry → alert from the CURRENT status; Adjust changes only that piece	Pass
API-PR31 BVA: planned at 24 °C, now 30 °C → weather alert on the wool blazer; 28 °C → no alert	Pass
API-PR32 Packed and Borrowed planned pieces are caught too	Pass

Borrow tracker (1)

Test	Result
API-PR40 borrower and dates persist; overdue reminder; Remind me later; Mark returned → Available	Pass

Cost per wear (1)

Test	Result
API-PR50 optional price; recalculated after a confirmed wear and after an edit; removable	Pass

Outfit memory and saved looks (3)

Test	Result
API-PR60 wearing stores the context (event, temperature, rating); Ask Closet finds it	Pass
API-PR61 Save then Favourite (and Favourite then Save) keep ONE record (defect D-10)	Pass
API-PR62 repeat insight: a 5★ look is offered again; with repeats switched off a same-week copy suggests a variation	Pass

Smart shopping: duplicates, gap confidence, purchase impact (3)

Test	Result
API-PR70 adding a near-duplicate warns first; the user can still add it	Pass
API-PR71 purchase simulation uses the real wardrobe and never saves the piece	Pass
API-PR72 every gap has a confidence and reasons; HIGH comes first	Pass

Packing Mode trip capsule (1)

Test	Result
API-PR80 carry-on limits, reuse, and Mark as packed changes the REAL item status (no duplicate records)	Pass

Ask Closet routes to real state (several different intents) (4)

Test	Result
API-PR90 “Is my blazer clean?” reads state and changes nothing; “Put the blazer in the wash” changes it	Pass
API-PR91 “What could I donate?” → the denim jacket; “I want to keep it” snoozes THAT piece	Pass
API-PR92 “Mark the denim jacket for donation” then “I donated the denim jacket” → gone, no replacement	Pass
API-PR93 different questions get different functions and answers	Pass

Style quiz and profile (US-02, US-03, US-04) (5)

Test	Result
API-P01 new profile starts empty with quiz not done	Pass
API-P02 quiz answers are saved and the quiz is marked done	Pass
API-P03 the quiz can be skipped (empty answers) and is not shown again	Pass
API-P04 settings change dress code, modest option and city later	Pass
API-P05 dress code 6 and unknown style → 400	Pass

Planner (US-46) (5)

Test	Result
API-E01 add an event: title, date, time, one occasion; dress code defaults from occasion	Pass
API-E02 month view returns only that month, in date order	Pass
API-E03 impossible date and time → 400	Pass
API-E04 bad month parameter → 400	Pass
API-E05 edit and delete my event; another user's event is not found	Pass

Closet gaps (US-64) (1)

Test	Result
API-G01 gaps follow the levels the user dresses for	Pass

Delete my account (US-06) (3)

Test	Result
API-D01 without the confirmation phrase → 400, nothing deleted	Pass
API-D02 wrong password → 401, nothing deleted	Pass
API-D03 confirmed → every row and every photo of the user is gone; token stops working	Pass

Security pipeline (5)

Test	Result
API-X01 security headers are set and the server is not advertised	Pass
API-X02 CORS allows only the Closet web app	Pass
API-X03 broken JSON → 400 with a safe message	Pass

Test	Result
API-X04 unknown endpoint → 404 JSON	Pass
API-X05 an internal error returns a generic 500 without details	Pass

Core outfit flow: 3 outfits → change a piece → save → wear → feedback (UC-09, UC-12) (2)

Test	Result
API-CF01 three complete outfits, then change one piece: only that slot changes and the look still passes the rules	Pass
API-CF02 save as favourite, wear with a rating, reject another look with a reason — each is stored as feedback	Pass

Deletion reasons drive shopping and learning (US-13, US-33, US-34) (5)

Test	Result
API-DR01 Torn / damaged → the replacement is on the shopping list at once, and the reply says so	Pass
API-DR02 Became too small → a replacement in the next size	Pass
API-DR03 No longer my style → nothing on the list, never suggested, and a buy check says think twice	Pass
API-DR04 two streetwear pieces let go → “Closet has learned”; confirm avoids streetwear in outfits; Undo restores the profile exactly	Pass
API-DR05 the reason is required and must be one of the four (old reasons are refused); Other changes nothing	Pass

Shopping list and Ask Closet (US-36, US-65, US-88) (5)

Test	Result
API-SL01 “Put beige trousers on my shopping list” → on the list, “Added by Ask Closet”, with a similar-piece warning	Pass
API-SL02 after “Do I need …?”, “add this to my shopping list” adds that item; asking twice does not duplicate it	Pass
API-SL03 “Add a replacement for my torn loafers” → tagged Replacement and Added by Ask Closet, no duplicate warning	Pass
API-SL04 Undo removes what Ask Closet added; mark bought, remove; another user cannot touch my list	Pass
API-SL05 a suggestion added to the bag leaves the suggestions; removing it from the bag brings it back	Pass

Add a piece (US-08, US-09, US-11) (10)

Test	Result
API-W01 photo + details → 201, saved once, status Available	Pass
API-W02 the photo is stored and shown through its signed URL	Pass
API-W03 location (EXIF) data is removed before the photo is stored	Pass
API-W04 PNG is accepted	Pass
API-W05 no photo → 400 and nothing saved	Pass
API-W06 a PDF renamed to .jpg → 415	Pass

Test	Result
API-W07 photo of exactly 10 MB is accepted (boundary)	Pass
API-W08 photo of 10 MB + 1 byte → 413 (boundary)	Pass
API-W09 missing category and no dress-code level → 400, photo not kept	Pass
API-W10 details that are not JSON → 400	Pass

Browse and filter (US-12) (3)

Test	Result
API-W11 all pieces with counts	Pass
API-W12 filters combine: Top AND Available	Pass
API-W13 invalid filter value → 400	Pass

Edit later (US-71) (2)

Test	Result
API-W14 edit changes the details and appears in filters	Pass
API-W15 invalid edit → 400 and nothing changes	Pass

Ownership and photo privacy (NFR-04, NFR-05) (3)

Test	Result
API-W16 another user cannot read, edit or change the status of my piece (404)	Pass
API-W17 photo URL with a changed signature or no signature → 404	Pass
API-W18 a malformed piece id → 404, not a server error	Pass

AI-40 56 natural-language wardrobe questions route to the right function (56)

Test	Result
“'What should I wear to university tomorrow?'” → 'What should I wear to university tomorrow?' #0	Pass
“'What can I wear to uni tomorrow?'” → 'What can I wear to uni tomorrow?' #1	Pass
“'What should I wear to dinner?'” → 'What should I wear to dinner?' #2	Pass
“'Style my navy blazer.'” → 'Style my navy blazer.' #3	Pass
“"What haven't I worn lately?"” → "What haven't I worn lately?" #4	Pass
“'Put my light wash jeans in the laundry.'” → 'Put my light wash jeans in the laundry.' #5	Pass
“'Put the blazer in the wash.'” → 'Put the blazer in the wash.' #6	Pass
“'Are my black loafers available?'” → 'Are my black loafers available?' #7	Pass
“'Is my blazer clean?'” → 'Is my blazer clean?' #8	Pass
“'Do I own something like a black blazer?'” → 'Do I own something like a black blazer?' #9	Pass
“'What is missing from my wardrobe?'” → 'What is missing from my wardrobe?' #10	Pass
“'Should I buy white sneakers?'” → 'Should I buy white sneakers?' #11	Pass

Test	Result
“'Do I need beige trousers?'” → 'Do I need beige trousers?' #12	Pass
“'Add this to my shopping list.'” → 'Add this to my shopping list.' #13	Pass
“'What is the weather tomorrow?'” → 'What is the weather tomorrow?' #14	Pass
“'Laundry is done.'” → 'Laundry is done.' #15	Pass
“'Show me pieces I forgot about.'” → 'Show me pieces I forgot about.' #16	Pass
“'What should I donate?'” → 'What should I donate?' #17	Pass
“'What clothes could I give away?'” → 'What clothes could I give away?' #18	Pass
“'Keep the denim jacket.'” → 'Keep the denim jacket.' #19	Pass
“'I want to keep it.'” → 'I want to keep it.' #20	Pass
“'Mark the denim jacket for donation.'” → 'Mark the denim jacket for donation.' #21	Pass
“'I donated the denim jacket.'” → 'I donated the denim jacket.' #22	Pass
“'Pack me for 5 days in Paris.'” → 'Pack me for 5 days in Paris.' #23	Pass
“'What should I bring to Milan next week?'” → 'What should I bring to Milan next week?' #24	Pass
“'Can I do this trip with a carry-on?'” → 'Can I do this trip with a carry-on?' #25	Pass
“'Do I need to do laundry before Monday?'” → 'Do I need to do laundry before Monday?' #26	Pass
“'Which clothes do I need to wash?'” → 'Which clothes do I need to wash?' #27	Pass
“'Can you plan my outfits for the next 4 days?'” → 'Can you plan my outfits for the next 4 days?' #28	Pass
“"What did I wear to Salma's birthday?"” → "What did I wear to Salma's birthday?" #29	Pass
“'What outfits did I like most?'” → 'What outfits did I like most?' #30	Pass
“"Can I repeat yesterday's outfit?"” → "Can I repeat yesterday's outfit?" #31	Pass
“'Why is my wardrobe health low?'” → 'Why is my wardrobe health low?' #32	Pass
“'What clothes should I pack away?'” → 'What clothes should I pack away?' #33	Pass
“'What should I bring back for summer?'” → 'What should I bring back for summer?' #34	Pass
“'Which clothes are out of season?'” → 'Which clothes are out of season?' #35	Pass
“'Who has my blazer?'” → 'Who has my blazer?' #36	Pass
“'What clothes have I lent out?'” → 'What clothes have I lent out?' #37	Pass
“'Mark the blazer returned.'” → 'Mark the blazer returned.' #38	Pass
“"Is tomorrow's outfit still okay for the weather?"” → "Is tomorrow's outfit still okay for the weather?" #39	Pass
“'I have a wedding tomorrow.'” → 'I have a wedding tomorrow.' #40	Pass
“'I have an interview in 2 hours.'” → 'I have an interview in 2 hours.' #41	Pass
“'I need something for dinner tonight.'” → 'I need something for dinner tonight.' #42	Pass
“'I have a wedding tomorrow and nothing to wear.'” → 'I have a wedding tomorrow and nothing to wear.' #43	Pass
“'What would these loafers add to my wardrobe?'” → 'What would these loafers add to my wardrobe?' #44	Pass
“'Would this blazer be useful?'” → 'Would this blazer be useful?' #45	Pass

Test	Result
“'How many outfits could I make with a white skirt?'” → 'How many outfits could I make with a white skirt?' #46	Pass
“'What has the best cost per wear?'” → 'What has the best cost per wear?' #47	Pass
“'Was my blazer worth it?'” → 'Was my blazer worth it?' #48	Pass
“'What are my best-value clothes?'” → 'What are my best-value clothes?' #49	Pass
“'What was my least-used expensive purchase?'” → 'What was my least-used expensive purchase?' #50	Pass
“'Do I actually need loafers?'” → 'Do I actually need loafers?' #51	Pass
“'Tell me a joke'” → 'Tell me a joke' #52	Pass
“'What can you do?'” → 'What can you do?' #53	Pass
“'Give me something similar to what I wore last Friday.'” → 'Give me something similar to what I wore last Friday.' #54	Pass
“'Should I buy black trousers?'” → 'Should I buy black trousers?' #55	Pass

Arguments that make the answer specific (7)

Test	Result
AI-41 “uni” is University; “tomorrow” is a date	Pass
AI-42 a garment word narrows the piece (“the blazer” is not every layer)	Pass
AI-43 availability questions never change state (they are not set_status)	Pass
AI-44 “this / it” follow-ups ask for the conversation context	Pass
AI-45 trips: destination, length and carry-on are read	Pass
AI-46 outside the wardrobe domain → a scoped answer, not a random outfit	Pass
AI-47 beige-trouser analysis only when beige trousers are asked about	Pass

Keyword parser: message → one whitelisted function (EP) (16)

Test	Result
'AI-01' “'AI-01'” → 'AI-01' #0	Pass
'AI-02' “'AI-02'” → 'AI-02' #1	Pass
'AI-03' “'AI-03'” → 'AI-03' #2	Pass
'AI-04' “'AI-04'” → 'AI-04' #3	Pass
'AI-05' “'AI-05'” → 'AI-05' #4	Pass
'AI-06' “'AI-06'” → 'AI-06' #5	Pass
'AI-07' “'AI-07'” → 'AI-07' #6	Pass
'AI-08' “'AI-08'” → 'AI-08' #7	Pass
'AI-09' “'AI-09'” → 'AI-09' #8	Pass
'AI-10' “'AI-10'” → 'AI-10' #9	Pass
'AI-11' “'AI-11'” → 'AI-11' #10	Pass

Test	Result
'AI-12' "'AI-12'" → 'AI-12' #11	Pass
AI-13 occasion, style and slot are extracted	Pass
AI-14 piece descriptions: synonyms and fabric	Pass
AI-15 dates: today, tomorrow, next weekday	Pass
AI-16 matching pieces by colour, category and fabric	Pass

Language Model Adapter (IIntentParser) (3)

Test	Result
AI-17 no API key → keyword parser	Pass
AI-18 with a key: the model's tool call is used (only whitelisted functions)	Pass
AI-19 model down → falls back to the keyword parser (NFR-06)	Pass

Weather Adapter (IWeather) (6)

Test	Result
WA-01 today uses the current temperature; source Open-Meteo	Pass
WA-02 a later day uses its maximum	Pass
WA-03 cached for 30 minutes: the second call does not hit the service	Pass
WA-04 service down → seasonal estimate, labelled (NFR-06)	Pass
WA-05 no city, or more than 16 days ahead → estimate	Pass
WA-06 demo mode (WEATHER_MODE=estimate): never calls the service; the same estimate every run, labelled	Pass

Version 1.2 intents and calendar import (9)

Test	Result
'AI-20' "'AI-20'" → 'AI-20' #0	Pass
'AI-21' "'AI-21'" → 'AI-21' #1	Pass
'AI-22' "'AI-22'" → 'AI-22' #2	Pass
'AI-23' "'AI-23'" → 'AI-23' #3	Pass
'AI-24' "'AI-24'" → 'AI-24' #4	Pass
'AI-25' "'AI-25'" → 'AI-25' #5	Pass
AI-26 bags and jewellery are recognised	Pass
AI-27 calendar titles → occasion	Pass
CAL-01 .ics: UTC time converted to Cairo, all-day events, escapes, folded lines	Pass

Closet gaps (7)

Test	Result
UT-G01 levels = usual dress code + each occasion default	Pass

Test	Result
UT-G02 0 pieces at a level is a gap	Pass
UT-G03 exactly 1 piece is still a gap (boundary)	Pass
UT-G04 2 pieces is not a gap (just above boundary)	Pass
UT-G05 levels the user never dresses for are not reported	Pass
UT-G06 a piece counts at every level it suits; message reads like the story example	Pass
UT-G07 Extra pieces are not part of any gap group	Pass

Diversity-aware ranking on the seeded closet (11)

Test	Result
DV-01 'University' at 'University' °C: exactly 3 looks, any two share at most ONE major garment #0	Pass
DV-01 'University' at 'University' °C: exactly 3 looks, any two share at most ONE major garment #1	Pass
DV-01 'University' at 'University' °C: exactly 3 looks, any two share at most ONE major garment #2	Pass
DV-01 'Work' at 'Work' °C: exactly 3 looks, any two share at most ONE major garment #3	Pass
DV-01 'Work' at 'Work' °C: exactly 3 looks, any two share at most ONE major garment #4	Pass
DV-01 'Dinner' at 'Dinner' °C: exactly 3 looks, any two share at most ONE major garment #5	Pass
DV-01 'Dinner' at 'Dinner' °C: exactly 3 looks, any two share at most ONE major garment #6	Pass
DV-02 the three University looks use three different tops and at least two different bottoms	Pass
DV-03 no two looks are the same set of pieces, and every look keeps a passing quality	Pass
DV-04 a slightly lower-ranked but different look beats a near-copy of the first look	Pass
DV-05 a tiny closet still returns looks (the constraint relaxes instead of failing)	Pass

Style this piece / Another chance keeps the chosen piece in every look (regression D-06) (5)

Test	Result
DV-06 'denim_jacket' for 'denim_jacket' at 'denim_jacket' °C: 3 looks, all contain it, other majors still diverse #0	Pass
DV-06 'camel_cardigan' for 'camel_cardigan' at 'camel_cardigan' °C: 3 looks, all contain it, other majors still diverse #1	Pass
DV-06 'denim_jacket' for 'denim_jacket' at 'denim_jacket' °C: 3 looks, all contain it, other majors still diverse #2	Pass
DV-06 'cream_knit_sweater' for 'cream_knit_sweater' at 'cream_knit_sweater' °C: 3 looks, all contain it, other majors still diverse #3	Pass
DV-07 a piece in the laundry cannot be styled: no looks and a clear note	Pass

Filter step (4)

Test	Result
RE-01 unavailable pieces are never used (US-17)	Pass
RE-02 every main piece suits the occasion level (US-21)	Pass

Test	Result
RE-03 modest on → no non-modest piece appears (US-21)	Pass
RE-04 hidden pieces are skipped for 7 days (US-61)	Pass

Weather rules (BVA at 19 and 26 °C, US-20) (4)

Test	Result
RE-05 25 °C: warm pieces allowed	Pass
RE-06 26 °C: no warm piece at all	Pass
RE-07 18 °C: every outfit has a warm layer	Pass
RE-08 19 °C: no layer required	Pass

Combine and score (8)

Test	Result
RE-09 three outfits, each complete, no two share more than 2 pieces (US-19)	Pass
RE-10 rule score = 100 − 17 per failed criterion (§3.1)	Pass
RE-11 colour harmony: red + pink clash; 3 accents fail; neutrals pass	Pass
RE-12 every outfit has 3–5 reasons and a rule score (US-57)	Pass
RE-13 “Style this piece”: all 3 outfits contain it (US-59)	Pass
RE-14 a forgotten piece is ranked up and named in the reasons (US-54)	Pass
RE-15 not enough pieces → no outfit and a clear note	Pass
RE-16 a look rejected as “Not in the mood today” is not suggested again this week	Pass

Change one piece and the outfit builder (4)

Test	Result
RE-17 swap changes only that slot, with an Available alternative (US-24)	Pass
RE-18 swap with no alternative returns null	Pass
RE-19 builder: pass/fail per criterion and one sentence naming the failure (US-43, US-66)	Pass
RE-20 builder: incomplete outfit fails “Complete outfit”	Pass

Version 1.2: recycle time, bags, Mix & Match breakdown, titles (9)

Test	Result
RE-21 recycle time: a top worn 1 day ago waits when repeatDays = 3	Pass
RE-22 recycle time BVA: worn exactly N days ago is allowed again	Pass
RE-23 recycle time never blocks shoes, bags or accessories	Pass
RE-24 a bag fills the fourth slot and must suit the level	Pass
RE-25 breakdown: all-pass look scores high; a clash drops colour to 40	Pass
RE-26 breakdown: one level off gives half credit for formality	Pass

Test	Result
RE-27 breakdown weather: warm piece at 30 °C, no layer at 12 °C	Pass
RE-28 outfit title uses the level and the key piece name	Pass
RE-29 Mix & Match explanation names what to improve	Pass

Final pass — recommendation details (5)

Test	Result
RE-30 three looks that share pieces still get three different names	Pass
RE-31 Style this piece: a warm layer at 30 °C is still in every look	Pass
RE-32 Style this piece refuses a piece that is not Available	Pass
RE-34 same closet, new database ids → same looks and same tray order (a reseeded demo repeats itself)	Pass
RE-33 replace tray: with fewer than 3 exact-level pieces, pieces one level away are offered after them	Pass

Final pass — photo try-on preview placement (client/src/photoTryOn.js) (3)

Test	Result
TO-01 body order: top above bottom above shoes; shoes at the feet	Pass
TO-02 layering: outerwear is drawn over the top, the top over the bottom	Pass
TO-03 accessories go where they are worn: sunglasses at the head, a bag at the side	Pass

Long-unworn pieces: style it, keep it or donate it (user decides) (11)

Test	Result
IN-01 BVA: 90 days unworn → candidate; 89 days → not	Pass
IN-02 recently worn and newly added pieces are never candidates	Pass
IN-03 a 'Laundry' piece does not trigger (only Available pieces) #0	Pass
IN-03 a 'Borrowed' piece does not trigger (only Available pieces) #1	Pass
IN-03 a 'Packed' piece does not trigger (only Available pieces) #2	Pass
IN-03 a 'Unavailable' piece does not trigger (only Available pieces) #3	Pass
IN-04 a heavy winter piece is not judged after summer days (season context)	Pass
IN-05 special-occasion pieces get 180 days, everyday pieces 90	Pass
IN-06 Keep it snoozes the suggestion until the chosen date	Pass
IN-07 wearing the piece resets the clock; a piece being donated is not suggested again	Pass
IN-08 seeded closet: the denim jacket (104 days) is the candidate and the message is warm, not judgmental	Pass

Seasonal rotation (suggest, never automatic) (3)

Test	Result
IN-10 a summer-only piece becomes a candidate when it turns cold; a winter coat is not packed in the cold	Pass
IN-11 an already Packed piece is not suggested again; Bring back lists seasonal storage only	Pass

Test	Result
IN-12 a piece worn in the last 3 weeks is not packed away; Not now hides it	Pass

Repeat intelligence and outfit memory (5)

Test	Result
IN-30 an identical look within the week suggests a variation (unless repeats are fine)	Pass
IN-31 a highly rated look is offered as a positive repeat	Pass
IN-32 a different look (only shoes shared) is not a repeat	Pass
IN-33 memory finds “Salma’s birthday” by its words, best-rated looks, and last Friday	Pass
IN-34 memory is one bounded signal: +1 per rated look, never more than +3, same occasion ± 4 °C	Pass

Planned outfit vs weather (monitor) (3)

Test	Result
IN-40 BVA: +6 °C warns about the warm piece; +4 °C does not	Pass
IN-41 rain appearing warns even without a big temperature change	Pass
IN-42 no snapshot → no alert (the monitor never guesses)	Pass

Cost per wear (confirmed wears only) (3)

Test	Result
IN-50 price ÷ wears; zero wears never divides; no price works normally	Pass
IN-51 currency is shown as stored; an edit recalculates	Pass
IN-52 best value needs at least 3 wears; few prices give a cautious note	Pass

Gap confidence and purchase impact (4)

Test	Result
IN-60 HIGH: no neutral formal bottoms at the user’s main level, with tops waiting for them	Pass
IN-61 never HIGH when equivalents exist one level away; one piece at a secondary level is LOW/MEDIUM	Pass
IN-62 a missing useful piece creates many new combinations; a near-duplicate is “Probably unnecessary”	Pass
IN-63 the simulation never adds the piece and counts every OWNED piece (laundry does not distort it)	Pass

Trip capsule (Packing Mode) (4)

Test	Result
IN-70 carry-on: at most 8 clothing pieces and 2 pairs of shoes; pieces are reused	Pass
IN-71 unavailable pieces are never packed; no physical piece appears twice	Pass
IN-72 the weather changes the choices (cold trip → a warm layer)	Pass
IN-73 a top is worn at most twice without laundry, three times with it; Remove excludes a piece	Pass

Wardrobe Health is explainable (1)

Test	Result
IN-80 the same wardrobe gives the same score, and the explanation names what lowers it	Pass

Explainable recommendations (US-57, v1.3) (6)

Test	Result
EX-01 every outfit has "why" factors and a "Recommended because" sentence	Pass
EX-02 the explanation names the real context: occasion, dress code, weather and city	Pass
EX-03 availability, favourites and rejection patterns appear when they shaped the look	Pass
EX-04 a piece in Laundry or Packed is never recommended (conflict rule)	Pass
EX-05 "Too formal" feedback makes relaxed pieces rank higher, never changing the rule score	Pass
EX-06 explanationSentence picks at most three factors and ends with a full stop	Pass

Ask Closet routes the duplicate question (US-65) (1)

Test	Result
DU-03 “Do I need another white top?” → buy_check, not the general shopping list	Pass

Packing planner and occasion templates (4)

Test	Result
PK-01 one outfit per day, pieces re-used, counts add up	Pass
PK-02 never packs an unavailable piece; rain with no coat → waterproof warning	Pass
PK-03 a day with no possible outfit is reported as missing	Pass
OT-01 nine occasion templates, each mapped to a real occasion, level and style	Pass

Offline demo = real services on in-memory data (9)

Test	Result
OF-01 seeded like npm run seed: 30 pieces, 3 in the laundry, the denim jacket forgotten for 104 days	Pass
OF-02 without the session token every protected call is 401	Pass
OF-03 Laundry propagates everywhere and Wash done brings it back	Pass
OF-04 an invalid transition is refused by the same state machine (409)	Pass
OF-05 Favourite and Save reach ONE record, visible in Favourites	Pass
OF-06 Ask Closet: different questions, different functions, follow-up “this” updates the bag	Pass
OF-07 Donate: mark → done → gone from the closet, no replacement, history kept	Pass
OF-08 state survives a page refresh in the same tab (serialize → restore)	Pass
OF-09 the planned-outfit monitor sees the warmer forecast on the seeded presentation outfit	Pass

Login tokens (JWT) (5)

Test	Result
UT-S01 a fresh token is accepted and carries id, role, version	Pass

Test	Result
UT-S02 a token signed with another secret is refused	Pass
UT-S03 an expired token is refused	Pass
UT-S04 the "none" algorithm is refused	Pass
UT-S05 garbage is refused	Pass

Passwords (bcrypt) (2)

Test	Result
UT-S06 hash is not the password and verifies only the right password	Pass
UT-S07 an account without a password (Google only) never matches	Pass

Photos (8)

Test	Result
'UT-P01 JPEG' #0	Pass
'UT-P02 PNG' #1	Pass
'UT-P03 WebP' #2	Pass
'UT-P04 HEIC' #3	Pass
'UT-P05 PDF is refused' #4	Pass
'UT-P06 EXE is refused' #5	Pass
UT-P07 EXIF (GPS) block is removed, the image data is kept	Pass
UT-P08 signed URL: valid, expired, tampered, foreign key	Pass

Access control and pipeline (2)

Test	Result
UT-S08 admin route: admin passes, user gets 403, no user gets 401	Pass
UT-S09 HTTP is redirected to HTTPS in production only	Pass

Event bus (Observer) (1)

Test	Result
UT-E01 every subscriber receives the event, in order; unknown topics are ignored	Pass

Deletion reasons (US-13) (4)

Test	Result
DR-01 exactly four reasons, each with a defined effect; only Torn / damaged and Became too small ask for a replacement	Pass
DR-02 replacementFor: torn → same piece; too small → next size; no longer my style and other → nothing	Pass
DR-03 US-34: a suggestion is avoided only when category AND colour match a “No longer my style” deletion	Pass
DR-04 a style is "let go" only after two or more “No longer my style” deletions share it (boundary 1 / 2)	Pass

Learning from deletions (US-34 → UC-22) (2)

Test	Result
DR-05 two let-go streetwear pieces → a pattern with evidence; confirming avoids the style and removes it from the profile; Undo keeps the old values	Pass
DR-06 the ranker lowers pieces in an avoided style (rank only — the rule score is not involved)	Pass

Ask Closet → shopping list (US-36) (2)

Test	Result
SL-01 “Put beige trousers on my shopping list” → add_to_shopping with the piece and the user’s word	Pass
SL-02 “add this to my shopping list” uses the last item; “Add a replacement for my torn loafers” is a replacement	Pass

Item status state machine (28)

Test	Result
ST-01 a new piece starts as Available	Pass
ST-02 'Available' → 'Available' follows the state diagram #0	Pass
ST-02 'Available' → 'Available' follows the state diagram #1	Pass
ST-02 'Available' → 'Available' follows the state diagram #2	Pass
ST-02 'Available' → 'Available' follows the state diagram #3	Pass
ST-02 'Available' → 'Available' follows the state diagram #4	Pass
ST-02 'Laundry' → 'Laundry' follows the state diagram #5	Pass
ST-02 'Laundry' → 'Laundry' follows the state diagram #6	Pass
ST-02 'Laundry' → 'Laundry' follows the state diagram #7	Pass
ST-02 'Laundry' → 'Laundry' follows the state diagram #8	Pass
ST-02 'Laundry' → 'Laundry' follows the state diagram #9	Pass
ST-02 'Borrowed' → 'Borrowed' follows the state diagram #10	Pass
ST-02 'Borrowed' → 'Borrowed' follows the state diagram #11	Pass
ST-02 'Borrowed' → 'Borrowed' follows the state diagram #12	Pass
ST-02 'Borrowed' → 'Borrowed' follows the state diagram #13	Pass
ST-02 'Borrowed' → 'Borrowed' follows the state diagram #14	Pass
ST-02 'Packed' → 'Packed' follows the state diagram #15	Pass
ST-02 'Packed' → 'Packed' follows the state diagram #16	Pass
ST-02 'Packed' → 'Packed' follows the state diagram #17	Pass
ST-02 'Packed' → 'Packed' follows the state diagram #18	Pass
ST-02 'Packed' → 'Packed' follows the state diagram #19	Pass
ST-02 'Unavailable' → 'Unavailable' follows the state diagram #20	Pass
ST-02 'Unavailable' → 'Unavailable' follows the state diagram #21	Pass

Test	Result
ST-02 'Unavailable' → 'Unavailable' follows the state diagram #22	Pass
ST-02 'Unavailable' → 'Unavailable' follows the state diagram #23	Pass
ST-02 'Unavailable' → 'Unavailable' follows the state diagram #24	Pass
ST-03 exactly 10 valid transitions	Pass
ST-04 unknown status cannot move	Pass

Password length (BVA, US-01) (5)

Test	Result
UT-V01 7 characters is rejected (just below minimum)	Pass
UT-V02 8 characters is accepted (minimum)	Pass
UT-V03 72 characters is accepted (maximum)	Pass
UT-V04 73 characters is rejected (just above maximum)	Pass
UT-V05 empty password is rejected	Pass

Email (EP, US-01) (5)

Test	Result
'UT-V06 valid email' #0	Pass
'UT-V07 missing @' #1	Pass
'UT-V08 missing domain ending' #2	Pass
'UT-V09 contains a space' #3	Pass
UT-V10 email is lower-cased and trimmed	Pass

Piece details (EP + BVA, US-09) (7)

Test	Result
UT-V11 valid details pass	Pass
UT-V12 category outside the list is rejected	Pass
UT-V13 missing colour is rejected	Pass
UT-V14 zero dress-code levels is rejected (below 1)	Pass
UT-V15 all five levels is accepted (maximum)	Pass
UT-V16 level 6 is rejected (outside 1–5)	Pass
UT-V17 size of 10 characters accepted, 11 rejected	Pass

Profile (EP + BVA, US-02 / US-03 / US-04) (7)

Test	Result
UT-V18 dress code %i accepted = 0 #0	Pass
UT-V18 dress code %i accepted = 1 #1	Pass

Test	Result
UT-V18 dress code %i accepted = 5 #2	Pass
UT-V18 dress code %i accepted = 6 #3	Pass
UT-V19 unknown style is rejected	Pass
UT-V20 city of 1 character rejected, 2 accepted	Pass
UT-V21 modest must be true or false	Pass

Events (EP + BVA, US-46) (12)

Test	Result
UT-V22 valid event passes	Pass
UT-V23 title of 80 characters accepted, 81 rejected	Pass
UT-V24 date '2026-02-28' valid = '2026-02-28' #0	Pass
UT-V24 date '2026-02-29' valid = '2026-02-29' #1	Pass
UT-V24 date '2028-02-29' valid = '2028-02-29' #2	Pass
UT-V24 date '2026-13-01' valid = '2026-13-01' #3	Pass
UT-V24 date '12/11/2026' valid = '12/11/2026' #4	Pass
UT-V25 time '00:00' valid = '00:00' #0	Pass
UT-V25 time '23:59' valid = '23:59' #1	Pass
UT-V25 time '24:00' valid = '24:00' #2	Pass
UT-V25 time '7:30' valid = '7:30' #3	Pass
UT-V26 occasion outside the list is rejected	Pass

Appendix B · Coverage by file

File / folder	Lines %	Branches %	Functions %	Uncovered lines
app.js	95.80	85.71	93.55	109-113
config.js	92.59	40.00	100.00	9-10
memoryRepositories.js	93.36	68.42	65.57	68-72 102-106 148-149 157-159
seedData.js	100.00	90.00	100.00	
eventBus.js	100.00	100.00	100.00	
accessControl.js	100.00	100.00	100.00	
audit.js	100.00	87.50	95.00	
authenticate.js	100.00	100.00	100.00	
errorHandler.js	92.86	80.00	100.00	17-18
security.js	83.33	66.67	75.00	38-46
activityModule.js	100.00	86.36	100.00	
adminModule.js	100.00	65.63	100.00	
assistantModule.js	83.33	60.17	80.25	59 105-107 122-125 170-173 205-208 211-2
keywordParser.js	100.00	100.00	100.00	
llmAdapter.js	97.75	70.97	87.50	74 77
availabilityModule.js	96.09	89.47	94.44	101 112 121-123 129-130
statusMachine.js	100.00	100.00	100.00	
identityRepository.js	100.00	100.00	100.00	
identityRoutes.js	100.00	93.33	100.00	
identityService.js	93.27	90.91	91.67	95-101
passwords.js	100.00	100.00	100.00	
tokens.js	100.00	100.00	100.00	
health.js	100.00	88.33	100.00	
insights.js	99.72	86.61	94.16	301
intelligenceModule.js	99.17	80.82	100.00	180-181
learning.js	93.07	83.72	94.74	38-44
proactive.js	99.49	74.11	90.00	78
outfitsModule.js	97.35	81.88	89.01	15-16 76-77 184-185 278 288
packingModule.js	100.00	76.70	94.55	
plannerModule.js	98.01	74.44	88.00	159-161 264 267 279
privacyModule.js	100.00	100.00	100.00	
profileModule.js	100.00	72.34	100.00	

File / folder	Lines %	Branches %	Functions %	Uncovered lines
engine.js	99.74	92.49	96.67	299
recommendationModule.js	99.15	89.71	97.06	108
weatherAdapter.js	100.00	88.46	83.33	
shoppingModule.js	99.68	82.86	94.57	305
closetGaps.js	100.00	92.31	100.00	
photoStorage.js	100.00	92.50	100.00	
wardrobeModule.js	98.58	86.36	98.31	175-177 264-265
deletionPolicy.js	100.00	91.67	90.91	
errors.js	100.00	100.00	100.00	
validate.js	98.98	97.62	100.00	58
vocabulary.js	100.00	100.00	100.00	
all files	97.24	83.42	91.09	